Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/31/2023
PI Name:	Patla, Bijunath Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80305	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	New Principal Investigator as of April 1, 2021 is Dr. Bijunath R. Patla; previous PI was Dr. Chris Oates.		
COI Name (Institution):			
Grant/Contract No.:	NNH12AT81I		
Performance Goal No.:			
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NOTE: Continuation of "NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado (PI=Oates)" with new PI, Dr. Bijunath R. Patla.
To support primary Atomic Clock Ensemble in Space (ACES) mission objectives, the National Institute of Standards and Technology (NIST) proposes to:

1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10^-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA (formerly known as the Joint Institute for Laboratory Astrophysics); and to support general ACES science goals.

2. Participate as fully as possible in coordinated European Space Agency (ESA) data analysis with formal ESA Data Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to

Task Description:

define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources.

As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following:

• Global time and frequency transfer with uncertainty as low as 10^-17.

• Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10^-17 or better.

• Test of gravitation redshift at 2 x 10^-6 uncertainty or better.

• Search for variation in fundamental constants, such as fine structure constant at level of 10^-18 per year or better.

All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list):

• World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10^-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10^-16 or better.

• Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 9 x 10^-19.

• Two ytterbium lattice optical clocks with uncertainties better than 2 x 10^-18.

• Strontium lattice optical clocks with uncertainties of 2 x 10^-18 (at JILA). A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.

• A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons with demonstrated instability below 10^-16.

• A world-class time scale comprising eight hydrogen masers and a collection of Cs clocks.

NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications.

In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment.

We propose the following revised program plan outline:

Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction.

Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure.

Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time.

Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Gap Year (2017): Perform relativistic calculations to prevent unanticipated shifts from contaminating ACES data. Install updated MWL hardware into new NIST time and frequency transfer system. Prepare updated data analysis system in anticipation of ACES launch. $150k for partial support of staff time.

Gap 6 Mo. (2018): Develop and install precision time links between the MWL hardware site and NIST clocks. $80k for partial support of staff time.

Gap 12 Mo. (2018-2019): Assuming successful launch in 2020, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Develop a frequency-based time scale for ACES frequency comparisons. $80k for partial support of staff time.

Gap 12 Mo. (2019-2020): Assuming successful launch in 2021, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Evaluate and improve the performance of the new frequency-based time scale optimized for ACES frequency comparisons. $80k for partial support of staff time.

Gap 12 Mo. (2020-2021): Evaluate and improve the performance of the new frequency-based time scale. Upgrade hardware infrastructure to add resiliency and robustness to the frequency-based time scale. $80k for partial support of staff time.

Gap 13 Mo. (2021-2022):Develop and integrate methods for precise orbit determination. Accurately estimate higher-order relativistic effects,estimate time and frequency transfer noise for clocks in orbit. $90k for partial support of staff time.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The major benefits and impacts of this research program can be grouped into three main areas:
1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements.

2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory.

3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such as search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.

Task Progress:

In the most recent reporting period (February 2022 through January 2023), work has focused on three main tasks:
1. Build on the theoretical framework for estimating higher order relativistic effects associated with frequency transfer, and estimating fractional frequency shifts due to Earth's gravitational potential.

2. Develop and implement algorithms for generating continuous and low-noise frequency outputs for use in fundamental physics applications.

3.Upgrade and integrate hardware for synthesizing reference frequencies for investigating fundamental physics using clocks in space.

Progress on each task is described below:

1. Build on the theoretical framework for estimating higher order relativistic effects associated with frequency transfer, and estimating fractional frequency shifts due to Earth's gravitational potential.

The associated gravitational effects of comparing atomic clocks separated by short and long distances using two of the very latest geoid models was investigated. This is relevant as the performance and stability of clocks continue to improve. The equipotential surface of the Earth coinciding with the mean sea level – as determined by the least squares fit – is known as the geoid. The uncertainty in the estimation of the shape of the geoid affects the fractional frequency estimates of clocks located at different locations. NIST has collaborated with National Geodetic Survey (NGS) and estimated the geopotential uncertainties at various locations around NIST with respect to the NGA station Q407 in 2017. The critical point is that the uncertainty in estimating the height from the geoid is assumed to be zero.

Comparing clocks located far away is more challenging if the height from the geoid is not known to be within less than a centimeter. We continue to develop theoretical frameworks for integrating space-based high-precision clocks for applications in geodesy that can first reverse this problem of estimating the geoid more precisely using "Chronometric leveling".

We refined our studies of frequency transfer from a rotating, almost freely falling satellite such as the International Space Station (ISS), to a ground-based receiver. This involves several reference frames: the Earth-Centered Inertial (ECI) frame; the Earth-Centered Earth-Fixed (ECEF) frame; the non-rotating, almost freely-falling frame attached to the center of mass of the satellite. We also introduce a reference frame that is tied to the Keplerian orbit of the satellite, related to the frame by a rigid rotation, and a body-fixed frame.

Transformations between these coordinate frames are essential in developing all the gravitational and non-gravitational effects on photons propagating from a rotating satellite to a ground-based receiver. We derive transformations between these coordinate systems and calculate the transformation coefficients – that are applied to calculate the metric tensors in the different coordinate systems. The coordinate frequencies can be converted to the proper frequency using the components of the metric tensors.

We conclude that the gravitational potential difference between transmitter and receiver, together with several time-dilation contributions due to motion and rotation of the satellite and Earth rotation, determine the frequency shift. Several first-order Doppler contributions were evaluated precisely.

2. Develop and implement algorithms for generating continuous and low-noise frequency outputs for use in fundamental physics applications.

We continue to develop and update our Kalman-like filter by adding several numerical routines to accommodate the available hourly frequency offset updates made available by a continuously running fountain and occasional calibration from optical clocks. [Ed. Note: See References.]

When fully integrated, we hope to have the capability to compare distant clocks with a better understanding of the uncertainty budget. That means improved dead time uncertainty estimation, improved uncertainty estimates due to drifts, and other environmental effects. All the above would help deliver better outcomes for fundamental physics experiments.

Now we can predict the frequency and time of 20 different clocks to within a few seconds – using dynamic time and frequency noise for each clock averaged for ~10 days. The dispersion in time prediction errors over 30 days for well-behaving H-masers was under 20 ps, and for the ensemble – within 10 ps. The stability requirement for orbiting

	clocks, including ACES, is under 100 ps/day. Improvements to our Kalman-like filter include capability of including dynamically updated measurement and clock noise for each computation. Our frequency-scale algorithm and Kalman-like filtering scheme for obtaining a steered output that is smooth and continuous in frequency compared to the steered outputs of traditional timescales. For applications involving clock comparison tests of General Relativity (GR) and other tests of fundamental physics, estimated uncertainties in the observables depend on the frequency noise. Here we use a single H-maser that is the input to our steered clock for demonstrating the advantages of the frequency scale compared to a traditional (NIST) timescale. 3.Upgrade and integrate hardware for synthesizing reference frequencies for investigating fundamental physics using clocks in space. We integrated a continuously running Cs-fountain for generating a more robust, stable, and accurate frequency output. In order to transfer this signal to different buildings onsite, we upgraded the coaxial cable links and fiber links connecting different buildings. Recently we compared year-long data for the timing signals as seen in two different NIST buildings, B1 and B81. The UTC (NIST) pulse-per-second is synthesized in building B1. B1 signal (frequency) is transported via coaxial cables, and the pulse is re-generated at B81. This requirement is because the ACES terminal is to be in B81. References Hoth GW, Patla B, Ashby N, Gerginov V. "Initial Study of the Distributed Cavity Phase Shift for the New Microwave Cavities of Cs Fountains at NIST." Proceedings. 2022 Joint Conference of the European Frequency and Time Forum and IEEE International Frequency Control Symposium (EFTF/IFCS), Paris, France, April 24-28, 2022. http://dx.doi.org/
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